MCL-Team: Minimal Multi-Model Collaboration Surpasses Single Models on LiveBench and IFEval

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Title & Author Information

- **Title**: MCL-Team: Minimal Multi-Model Collaboration Surpasses Single Models on LiveBench and IFEval
- **Author**: Li Zhang
- **Affiliation**: TuringCorp

- **Version**: v1 (preprint)

- **Disclosure Boundary**: This report discloses the high-level algorithmic pipeline, benchmark methodology, and all evaluation results. Proprietary prompts, model routing configurations, and implementation-level workflow controls remain unpublished.
- **Repositories**: LaTeX source: https://github.com/TuringCorp-net/mcl-team-paper-preprint. Benchmark result packages, model cards, and evaluation methodology: https://github.com/TuringCorp-net/turingcorp-llm.

Abstract

We present MCL-Team, a multi-model collaboration algorithm that operationalizes the Meta-Coordination Layer (MCL) framework [1] for public benchmark evaluation. The algorithm applies MCL coordination mechanisms—including heterogeneous parallel generation, cross-optimization, and fusion-based selection—across three quality-cost tiers (Junior, Senior, Principal). On the LiveBench 2026-01 leaderboard, MCL-Team produces Language (93.5–94.2) and Instruction Following (87.4–88.8) scores exceeding all single models; Reasoning climbs from 84.5 (Junior) to 97.0 (Principal). On IFEval, the algorithm achieves 92.6% prompt-level strict accuracy and 95.0% instruction-level strict accuracy, 5.3–8.9 points above tested single-model competitors. A clean ablation (Junior $\rightarrow$ Senior, identical resource allocation, increased coordination depth) yields +6.3 Reasoning points, isolating the contribution of the coordination mechanism from model quality. Multi-model collaboration increases token consumption by 5–10$\times$ and latency by 2–10$\times$ relative to a single call. Benchmark result packages, model cards, and an OpenAI-compatible API endpoint are available for independent verification.

Keywords: Multi-agent collaboration; Meta-Coordination Layer; LiveBench; IFEval; Multi-model; Benchmark

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1 Introduction

Single large language models face inherent constraints: fixed analytical patterns shaped by pretraining distributions, limited capacity for self-correction during generation, and reasoning gaps that cannot be fully resolved through scale alone. Multi-agent approaches have been proposed to address these constraints [2, 3, 4], yet most existing work remains at the framework level. Few multi-agent systems have been evaluated against leading single models on standardized, contamination-resistant public benchmarks under protocols that permit direct comparison.

The Meta-Coordination Layer (MCL) framework [1] introduced a formalism for agent collaboration: a task graph $G = (V, E)$ with role contracts, governance checks, and convergence gates. It proposed five collaboration algorithm families and provided initial human preference evidence ($n = 59$) that structured multi-agent organizations improve output quality over single-model baselines. However, the original MCL work did not evaluate any concrete algorithm on public benchmarks, leaving open the question of whether the framework's claimed benefits translate to standard, third-party performance measures.

This paper addresses that gap. We implement MCL-Team, the foundational algorithm in the cooperative cross-discussion family, and evaluate it on two public benchmarks—LiveBench (2026-01 leaderboard snapshot, 682 questions) and IFEval (541 prompts, 25 verifiable instruction types, programmatic scoring). Both benchmarks are designed to resist contamination: LiveBench refreshes questions monthly [5], and IFEval uses rule-based verification with no LLM judge [6].

We evaluate MCL-Team at three quality-cost operating points (Junior, Senior, Principal), configured through internal resource allocation and coordination depth. The Junior → Senior comparison serves as an ablation: identical resource allocation, differing only in coordination depth. We also submit the completed benchmark results to three LLMs for independent capability profiling: DeepSeek V4 Pro, Gemini Pro (latest), and ChatGPT 5.5 Pro are each given the same leaderboard data and asked to analyze the ranking and characterize the capability profile. All three independently converge on the same interpretation.

Our contributions are:

- **A minimal collaboration algorithm**: A compact application of MCL coordination mechanisms—heterogeneous parallel generation, cross-optimization, and fusion-based selection—that produces measurable quality gains over single-model baselines.
- **Standardized benchmark results**: Language (93.5–94.2) and IF (87.4–88.8) above all single models on the LiveBench leaderboard; Reasoning from 84.5 (Junior) to 90.8 (Senior) to 97.0 (Principal); IFEval prompt-level strict accuracy up to 92.6% and instruction-level strict accuracy up to 95.0%.

- **A clean ablation**: The Junior → Senior transition (identical resource allocation, increased coordination depth) yields +6.3 Reasoning and +4.9 Math, establishing that the coordination mechanism—not resource allocation—drives the gains.
- **Quantified cost structure**: Multi-model collaboration requires 5–10× tokens and 2–10× latency, defining a transparent cost-quality frontier.

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2 Related Work

2.1 Multi-Agent Frameworks

MetaGPT [2] encodes software engineering workflows as standardized operating procedures for role-playing agents. CAMEL [3] studies communicative agent behavior through role assignment in simulated societies. AutoGen [4] provides infrastructure for multi-agent conversations. Generative Agents [7] simulate human-like behaviors in interactive sandboxes. These works establish the viability of multi-agent architectures; their evaluations focus on case studies and qualitative analyses, creating an opportunity for standardized benchmark evaluation against leading single models.

2.2 MCL Conceptual Framework

The MCL framework [1] models agent organizations as task graphs $G = (V, E)$ with five algorithm families: cooperative, competitive, co-opetitive, cross-discussion, and iterative decision-making. The framework was validated through human preference studies (academia $n = 14$, industry $n = 45$) comparing MCL-organized agent teams against single-model baselines. The present work selects the foundational cooperative cross-discussion variant, instantiates it as the Team algorithm, and provides the first public benchmark evaluation of any MCL-based system.

2.3 Benchmark-Based Evaluation

LiveBench [5] is designed to resist contamination through monthly question refresh across diverse sources. Each monthly release provides a fixed leaderboard, enabling temporally fair comparison. IFEval [6] evaluates instruction following through 25 verifiable constraint types (length, format, keyword, language) with fully programmatic regex-based scoring. Unlike LLM-judged benchmarks, IFEval scores are deterministic and reproducible. Recent work on inference-time compute scaling [8, 9] shows that repeated sampling and majority voting can raise output quality. MCL-Team differs by applying structured internal coordination — division of labor, peer review, and coordinated fusion — rather than treating each model output as an independent sample.

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3 Method

MCL-Team is an engineering instantiation of the Meta-Coordination Layer framework [1]. It coordinates multiple heterogeneous language models through MCL mechanisms—including parallel generation with peer review, cross-optimization, and coordinated fusion—to produce a single consolidated output. The algorithm is provided as an OpenAI-compatible API endpoint (`/v1/chat/completions`); proprietary prompts, model routing configurations, and internal coordination parameters are not disclosed.

MCL-Team is offered at three quality-cost operating points (Table 1). Internal configuration details—including the specific coordination mechanisms, model selection, and parameter differences between tiers—are proprietary. The Junior → Senior comparison serves as an ablation: identical resource allocation, differing only in coordination depth. Any performance difference between the two is attributable to the coordination mechanism, independent of resource scaling.

Table 1: MCL-Team operating points.

Tier	**Configuration**	**Coordination**
Junior	Baseline	Standard
Senior	Baseline	Enhanced
Principal	Expanded	Enhanced

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4 Experimental Setup

4.1 Benchmarks

LiveBench (2026-01 leaderboard snapshot): 682 questions across 6 categories—Language, Instruction Following, Reasoning, Math, Data Analysis, and Coding. We use the standard evaluation protocol with zero modifications. All comparisons use the same monthly snapshot.

IFEval: 541 prompts covering 25 verifiable instruction types. Scoring is programmatic; each response either satisfies a constraint or it does not. We report four metrics: prompt-level strict/loose (whether *all* instructions in a prompt are satisfied) and instruction-level strict/loose (the fraction of *individual* instructions satisfied across all prompts).

4.2 Protocol

- **Framework**: lm-eval-harness v0.4.12, unmodified.
- **Endpoint**: An OpenAI-compatible chat completions API. Researchers may request an API key for independent reproduction; see https://github.com/TuringCorp-net/turingcorp-llm.

- **Reporting**: All runs are first and only runs for each tier-benchmark pair. Results are reported in full.

4.3 Baselines

LiveBench baselines are drawn from the public 2026-01 leaderboard: GPT-5.5 Thinking xHigh Effort, Claude 4.8 Opus xHigh, Gemini 3.5 Flash High, Claude Fable 5 xHigh, Qwen 3.7 Max. IFEval baselines are published scores for GPT-4o, Claude Opus 4, and Llama-3.1 405B.

4.4 Data Availability

All benchmark result packages (LiveBench and IFEval raw outputs for all three tiers), model cards, and evaluation methodology are available at https://github.com/TuringCorp-net/turingcorp-llm. The paper LaTeX source is available at https://github.com/TuringCorp-net/mcl-team-paper-preprint.

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5 Results

5.1 LiveBench

Table 2 reports the full LiveBench results.

Table 2: LiveBench (2026-01) results across tiers.

Tier	Overall	Lang.	IF	Reas.	Math	Data
Junior	80.5	93.5	87.4	84.5	75.5	61.5
Senior	84.0	94.2	88.8	90.8	80.4	65.8
Principal	**85.9**	94.0	86.9	**97.0**	85.0	66.5

Language and IF are already near-ceiling at Junior (93.5, 87.4) and improve marginally at higher tiers. Reasoning, Math, and Data Analysis scale more substantially. Principal achieves the highest Overall (85.9) and Reasoning (97.0) scores.

5.2 Comparison with Leading Single Models

Table 3 compares MCL-Team against top single models on the same LiveBench leaderboard.

Table 3: MCL-Team Principal vs. single models, LiveBench 2026-01.

Model	**Overall**	**Reas.**	**Lang.**	**IF**
MCL-Team Principal	**85.9**	**97.0**	94.0	86.9
GPT-5.5 Thinking xHigh	82.8	87.7	87.7	73.0
Claude 4.8 Opus xHigh	79.5	89.7	81.4	67.5
Gemini 3.5 Flash High	80.7	82.0	84.6	75.6
Claude Fable 5 xHigh	78.6	87.3	88.5	60.0

Across tiers, the best MCL-Team score exceeds the best single model in each dimension: Reasoning by 7.3 points (Principal 97.0 vs. Claude 4.8 Opus 89.7), Language by 5.7 points (Senior 94.2 vs. Claude Fable 5 88.5), and IF by 13.2 points (Senior 88.8 vs. Gemini 3.5 Flash 75.6).

5.3 IFEval

IFEval uses programmatic scoring with no LLM judge. Table 4 reports all four metrics across tiers; Table 5 compares against published single-model scores.

Table 4: IFEval results across all three tiers.

Tier	**Prompt-Strict**	**Prompt-Loose**	**Inst-Strict**	**Inst-Loose**
MCL-Team Junior	90.6%	92.6%	93.5%	95.0%
MCL-Team Senior	**92.6%**	**94.3%**	94.8%	**96.0%**
MCL-Team Principal	92.4%	93.9%	**95.0%**	**96.0%**

Table 5: IFEval vs. single-model competitors.

Model	**Prompt-Strict**	**Inst-Strict**
MCL-Team Senior	**92.6%**	94.8%
MCL-Team Principal	92.4%	**95.0%**
GPT-4o	83.7%	—
Claude Opus 4	87.3%	—
Llama-3.1 405B	85.6%	—

MCL-Team Senior achieves 92.6% prompt-strict accuracy, 5.3 points above Claude Opus 4 (87.3%), and 8.9 points above GPT-4o (83.7%). Principal reaches 95.0% instruction-strict accuracy, meaning 95% of individual instructions are followed correctly across all 541 prompts. All scores are fully reproducible; no LLM judge is involved.

5.4 Ablation: Coordination Depth

Junior and Senior share the same resource allocation; Senior applies additional coordination depth. This isolates the mechanism:

- **Reasoning**: 84.5 → 90.8 (+6.3)
- **Math**: 75.5 → 80.4 (+4.9)
- **Data Analysis**: 61.5 → 65.8 (+4.3)
- **Language / IF**: 93.5 → 94.2 (+0.7) / 87.4 → 88.8 (+1.4)

The pattern—substantial gains in reasoning-intensive dimensions, modest gains in already-high dimensions—is consistent with the coordination mechanism functioning primarily to correct model-specific reasoning gaps and blind spots.

5.5 Cost Structure

Multi-model collaboration increases token consumption by 5–10× and latency by 2–10× relative to a single model call. The cost structure is transparent: each operating point involves a fixed number of internal model calls determined by the tier configuration. The Principal tier occupies the upper end of both cost ranges.

5.6 Independent Capability Analysis

To obtain an impartial interpretation of the benchmark results, we submitted the LiveBench 2026-01 leaderboard data and MCL-Team Junior scores to three LLMs—DeepSeek V4 Pro, Gemini Pro (latest), and ChatGPT 5.5 Pro—with identical instructions: rank the models, identify anomalies, and characterize the capability profile. The LLMs did not participate in benchmark evaluation; they only analyzed the completed scores.

All three independently converged on the same interpretation: Junior's Language (93.5) and IF (87.4) rank above all single models, the Reasoning score (84.5) places in the upper tier but below the strongest reasoning-specialized models, and Math/Data Analysis are the primary constraints on overall score. All three further noted that the capability profile is structurally unusual: whereas most single models exhibit Reasoning $\geq$ Language, MCL-Team Junior shows Language $\gg$ Reasoning (93.5 vs. 84.5, a 9-point gap). This profile is rare among single models and, as the analysts noted, is consistent with a coordination- and verification-oriented architecture rather than a pure reasoning engine. The strength in Language and IF, combined with competitive but not dominant Reasoning, positions the system for planning-heavy, communication-intensive tasks—agent orchestration, enterprise consulting, complex task decomposition—rather than specialized math or coding benchmarks. ChatGPT explicitly characterized this as a "planning + verification" capability pattern. The only divergence across the three analyses was in emphasis (deployment implications vs. statistical rigor vs. architectural interpretation).

6 Discussion

6.1 Mechanism

The Junior → Senior ablation (+6.3 Reasoning, identical resources) isolates the contribution of coordination. The effect is consistent with three mutually reinforcing factors: heterogeneous pretraining distributions providing complementary knowledge coverage, diverse reasoning paths enabling mutual correction, and agreement or disagreement between models serving as a calibration signal for the fusion step.

6.2 Scope and Limitations

- **Math and Data Analysis**: MCL-Team's scores in these dimensions remain below leading single models in their respective specialized categories. The current pipeline does not integrate structured tool use (code execution, database queries), which likely explains the gap.
- **Coding**: The LiveBench Coding category was not evaluated. Whether multi-model collaboration improves or degrades code generation is an open question.
- **Implementation access**: Proprietary components (prompts, model routing, workflow controls) are not released. The algorithmic description is sufficient for understanding the approach, but does not enable exact reproduction.

6.3 Relationship to MCL

This paper provides the first public benchmark validation of the MCL hypothesis [1]. The Team algorithm is the foundational member of the cooperative cross-discussion family; its strong benchmark performance suggests that more complex human-team-like collaboration algorithms merit further investigation.

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7 Conclusion

We have presented MCL-Team, a minimal multi-model collaboration algorithm, and evaluated it on LiveBench and IFEval. The key findings:

- Language and IF scores (93.5–94.2, 87.4–88.8) exceed all single models on the LiveBench 2026-01 leaderboard.
- Reasoning scales from 84.5 to 97.0 across three quality-cost tiers. Increased coordination depth alone contributes +6.3 points, isolated via a clean ablation.
- IFEval prompt-strict accuracy reaches 92.6% (Senior) and instruction-strict accuracy reaches 95.0% (Principal), 5.3–8.9 points above tested single-model competitors.

- Collaboration increases token consumption (5–10×) and latency (2–10×), defining an explicit cost-quality frontier.

MCL-Team and single-model workflows serve complementary roles: the former for quality-sensitive, non-real-time tasks; the latter for cost-sensitive and latency-constrained applications. Benchmark result packages and model cards are publicly available at https://github.com/TuringCorp-net/turingcorp-llm.

Acknowledgments

We thank the developers of LiveBench and IFEval for maintaining contamination-resistant, programmatically scored benchmarks. The independent capability analyses by DeepSeek V4 Pro, Gemini Pro (latest), and ChatGPT 5.5 Pro provided useful external interpretation.

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